N. 1

$f(x) = x + \log(|x^2-3|)$

D: $x^2-3>0 \Rightarrow x > \pm\sqrt{3}$ $\quad x < -\sqrt{3} \lor x > \sqrt{3}$

Inoltre $f(x) = x + \log(x^2+3)$ per $x<-\sqrt{3} \lor x>\sqrt{3}$

$f(x) = x + \log(-x^2-3)$ per $-\sqrt{3}<x<\sqrt{3}$

Limiti

per $x \to -\infty$ $\quad x + \log(x^2-3) = x + 2\log(x) \Rightarrow -\infty$

$\lim_{x\to -\sqrt{3}^-} x + \log(x^2-3) \Rightarrow x + \log(0^+) = -\infty$

$\lim_{x\to \sqrt{3}^+} x + \log(x^2-3) \Rightarrow x + \log(0^+) = -\infty$

$\lim_{x\to +\infty} x + \log(x^2-3) \Rightarrow +\infty$

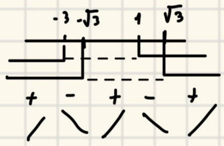
Derivata I

$f'(x) = (x + \log(x^2-3))' = 1 + \frac{1}{x^2-3} \cdot 2x = \frac{x^2-3+2x}{x^2-3}$

$x^2+2x-3>0 \Rightarrow x_{1,2} = \frac{-2\pm\sqrt{4+12}}{2} = \begin{cases} \frac{-2+4}{2} = 1 \\ \frac{-2-4}{2} = -3 \end{cases}$

$x>1$

$x>-3$



Derivata II

$f''(x) = \left(\frac{x^2+2x-3}{x^2-3}\right)' = \frac{(2x+2)(x^2-3)-(x^2+2x-3)(2x)}{(x^2-3)^2} =$

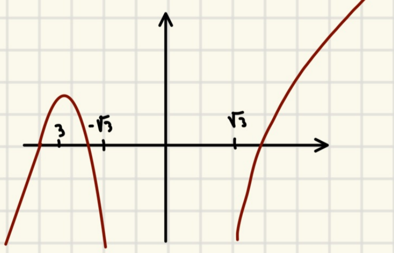
$\frac{\cancel{2x^3}-\cancel{6x}+2x^2-6-\cancel{2x^3}-4x^2+\cancel{6x}}{(x^2-3)^2} = \frac{-2x^2-6}{(x^2-3)^2} = \frac{-2(x^2+3)}{(x^2-3)^2}$

Non ci sono flessi, perché

$(x^2-3)^2>0 \quad x<-\sqrt{3} \lor x>\sqrt{3}$

$x^2+3>0 \quad \forall x\in\mathbb{R}$

$-2>0 \quad \forall x\in\mathbb{R}$



N. 2

$f(x) = \frac{1}{1+4\log(x)} - \frac{4}{x^{2x}+\sin(x^2-x)} = 1-4t+18t^2-4+12t-20t^2 = -3+8t-2t^2 = -3+8(x-1)-2(x-1)^2$

Poniamo $t = x-1$

$f(x) = \frac{1}{1+4\log(1+t)} - \frac{4}{(1+t)^{2+2t}+\sin((1+t)^2-(1+t))} =$

$(1+4\log(1+t))^{-1} = 1+-(4t-2t^2)+\frac{2}{2}(4t-2t^2) = 1-4t+2t^2+16t^2 = 1-4t+18t^2$

$(1+t)^{2+2t}+\sin(t^2+t) \quad = 1+2t+3t^2+t+t^2 = (1+3t+4t^2)^{-1}$

$4(1-3t-4t^2+9t^2)$

$4-12t+20t^2$

N. 3

$$f(x) = \frac{\arctan(\sqrt{6}x)}{x\sqrt{x}}$$

$$\int_0^{\infty} \frac{\arctan(\sqrt{6}\,x)x^{\alpha}\,dx}{x\sqrt{x}\,(\log(3+x^4))^2} = \int_0^{+\infty} \frac{\arctan(\sqrt{6}\,x)}{x^{3/2}(\log(3+x^4))^2}$$

I punti da indagare sono 0 e $+\infty$

Per $x \to 0^+$

$$\int_0^{+\infty} \frac{\arctan(\sqrt{6}\,x)x^{\alpha}}{x\sqrt{x}\,(\log(3+x^4))^2} \sim \frac{\sqrt{6}\sqrt{x}\,x^{\alpha}}{x\sqrt{x}\,(\log(3)^2)} = \frac{c}{x^{1-\alpha}}$$

l'integrale converge per $1-\alpha < 1$ ossia per $\alpha > 0$

Per $x \to +\infty$

$$\frac{\arctan(\sqrt{6}\,x)x^{\alpha}}{x\sqrt{x}\,(\log(3+x^4))^2} \sim \frac{\arctan(x)x^{\alpha}}{x\sqrt{x}(\log(x^4))^2} = \frac{\frac{\pi}{2}}{x^{\frac{3}{2}-\alpha}\log^2(x)}$$

l'integrale converge per $\frac{3}{2}-\alpha > 1$ ossia $\alpha < \frac{1}{2}$

Perciò l'integrale converge se e solo se $0 < \alpha < \frac{1}{2}$

b Calcolare $\int_{1/6}^{+\infty} f(x)\,dx$

$$\int_{1/6}^{+\infty} \frac{\arctan(\sqrt{6x})}{x\sqrt{x}} = \int_1^{+\infty} \frac{\arctan(t)}{\left(\frac{t^2}{6}\right)\left(\frac{t}{\sqrt{6}}\right)} \frac{dt\,t}{3} = \frac{1}{3}\int_1^{+\infty} \frac{6\sqrt{6}\arctan(t)}{t^3}\cdot t\,dt = \frac{6\sqrt{6}}{3}\int_1^{+\infty} \frac{\arctan(t)}{t^2}\,dt = 2\sqrt{6}\left[-\frac{\arctan(t)}{t} + \int \frac{1}{(1+t^2)t}\,dt\right]_1^{+\infty} =$$

$$\sqrt{6x} = t$$

$$dx\,\frac{3}{\sqrt{6x}} = dt$$

$f = \arctan(t)$ $\quad f' = \frac{1}{1+t^2}$

$g = -\frac{1}{t}$ $\quad g' = t^{-2}$

$$\frac{A}{t} + \frac{Bt+C}{1+t^2} = Bt^2Ct + A + At^2 : \begin{cases} A = 1 \\ C = 0 \\ B = -1 \end{cases}$$

$$2\sqrt{6}\left[-\frac{\arctan(t)}{t}\right]_1^{+\infty} + \int_1^{+\infty} \frac{1}{t} - \frac{t}{1+t^2}\,dt =$$

$$2\sqrt{6}\left[-\frac{\arctan(t)}{t} + \ln(t)\right]\frac{1}{2}\int_1 -\frac{2t}{1+t^2}\,dt = 2\sqrt{6}\left[-\frac{\arctan(t)}{t} + \ln(t) - \frac{1}{2}\ln(1+t^2)\right]_1^{+\infty} =$$

$$2\sqrt{6}\left[\frac{\pi}{4} + \left[\ln\left(\frac{t}{\sqrt{1+t^2}}\right)\right]_1^{+\infty}\right] = \sqrt{6}\,\frac{\pi}{2} - \ln\left(\frac{1}{\sqrt{2}}\right) = \sqrt{6}\left(\frac{\pi}{2} + \ln(2)\right)$$

N. 4

$$\cos(x)\,y'(x) = \sin(x)y(x) + e^{2x}(x-3)$$

$$y'(x) - \frac{\sin(x)\,y(x)}{\cos(x)} = \frac{e^{2x}(x-3)}{\cos(x)}$$

troviamo una primitiva per $-\tan(x)$

$$A(x) = \int \frac{\sin(x)}{\cos(x)}\,dx = \int \frac{dt}{t} = \ln(\cos(x))$$

$\cos(x) = t$, $dt = -\sin(x)\,dx$

Il fattore integrante è $e^{A(x)} = e^{\ln(\cos(x))} = \cos(x)$

$$\int e^{A(x)} \cdot f(x) = \int e^{2x}(x-3) = \frac{e^{2x}(x-3)}{2} - \int \frac{e^{2x}}{2} = \frac{e^{2x}(x-3)}{2} - \frac{1}{4}e^{2x} = \frac{e^{2x}}{2}\left[x - \frac{1}{2} - 3\right] = \left(\frac{e^{2x}}{2}\left[x - \frac{7}{2}\right] + c\right)$$

$g = \frac{e^{2x}}{2}$ $\quad g' = e^{2x}$

$f = (x-3)$ $\quad f' = 1$

Poniamo la condizione iniziale

$$y(0) = 2 = \frac{1}{\cos(x)}\left(-\frac{7}{4} + c\right) \Rightarrow c = \frac{7}{4} + 2 \quad = c = \frac{15}{4}$$

Per cui la soluzione è

$$y(x) = \frac{1}{\cos(x)}\left(\frac{e^{2x}}{2}\left[x - \frac{7}{2}\right] + \frac{15}{4}\right)$$

$y(0) = 2$

$y'(0) = -3$

$$y = y'(0)(x-0) + y(0) = -3x + 2$$